Departement Elektriese, Elektroniese en Rekenaar-Ingenieurswese

Semestertoets 1

Module EBN122

Elektrisiteit en Elektronika

27 Augustus 2015

Department of Electrical, Electronic and Computer Engineering

Semester Test 1

Module EBN122

Electricity and Electronics

27 August 2015

UNIVERSITEIT VAN PRETORIA
UNIVERSITY OF PRETORIA
YUNIBESITHI YA PRETORIA

Eksamensinligting: ***Exam information:***	Memo		
Maksimum punte: *Maximum marks:*	45	Volpunte: *Full marks:*	45
Duur van vraestel: *Duration of paper:*	90 minute *90 minutes*	Oopboek / toeboek: *Open / closed book:*	Toe *Closed*
Totale aantal bladsye (hierdie blad ingesluit): *Total numer of pages (including this page):*		14	

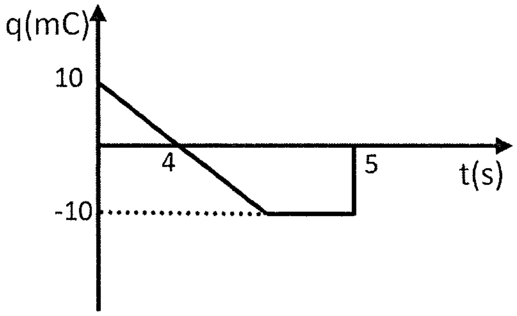
BELANGRIK- ***IMPORTANT***

1. Die eksamenregulasies van die Universiteit van Pretoria geld.
   The examination regulations of the University of Pretoria apply.
2. Alle vrae moet op die ANTWOORDBLAD beantwoord word.
   All questions must be answered on the ANSWER SHEET.
3. Studente mag hul eie rowwe berekeninge op die vraestel doen – die vraestel word nie ingeneem nie.
   Students may do their own rough calculations on the question paper – the question paper will not be taken in.
4. Antwoorde moet eenhede insluit!
   Answers must include units!
5. **Finale antwoorde moet as reële getalle geskryf word, en nie as breuke nie.**
   Final answers must be written as real numbers, and not as fractions.

Dosent(e): *Lecturer(s):*	1 J Joubert 2 A Chatterjee

VRAAG/QUESTION 1 [22]

Vraag 1.1 [2]	**Question 1.1 [2]**
Die lading deur 'n element word hieronder getoon. Bepaal die stroom by t = 4s (in mA).	The charge through an element is shown below. Find the current at t = 4s (in mA).



$$i = \frac{dq}{dt} = \frac{-10\times10^{-3}}{4}$$

$$= \underline{-2.5\ mA}$$

Vraag 1.2 [2]	**Question 1.2 [2]**
As 0.624×10^{18} elektrone deur 'n draad vloei in 5s, bepaal die stroom (in mA).	If 0.624×10^{18} electrons pass through a wire in 5s, find the current (in mA).

$$1e^- \rightarrow 1.6\times10^{-19}\ C$$

$$0.624\times10^{18} \xrightarrow{\times} x$$

$$x = 0.1\ C$$

$$\therefore\ i = \frac{q}{t}$$

$$= \frac{0.1}{5} = \underline{20\ mA}$$

Vraag 1.3 [2]	Question 1.3 [2]
'n 15Ω weerstand is gekoppel aan 'n 10V battery. Hoeveel energie (in Joule) sal in die weerstand verkwis word in 1 minuut?	A 15Ω resistor is connected to a 10V battery. How much energy (in Joule) will be dissipated in the resistor in 1 minute?

$$E = P \times t$$
$$= \frac{V^2}{R} \times t$$
$$= \frac{10^2}{15} \times 60$$
$$= \underline{\underline{400J}}$$

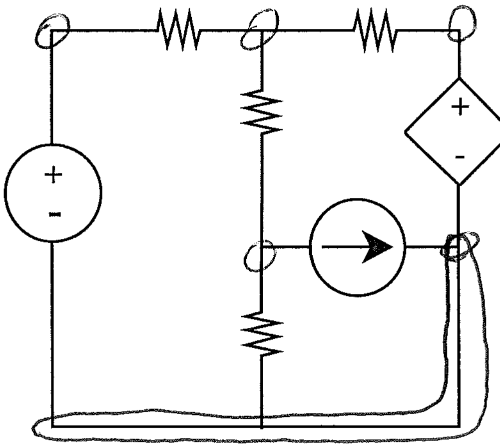
✓✓

Vraag 1.4 [2]	Question 1.4 [2]
Vir hoeveel ure kan jy 'n 250W TV gebruik vir 'n totale koste van 75 sent, as die koste van elektrisiteit R1/kWh is?	At electricity cost of R1/kWh, how many hours can you watch a 250W TV for a total cost of 75 cents?

$$100c \longrightarrow 1000 \, Wh$$
$$75c \longrightarrow x$$
$$\therefore x = 750 \, Wh$$
$$\therefore \text{time} = \frac{750 \, Wh}{250 \, W}$$
$$= \underline{\underline{3h}}$$

✓✓

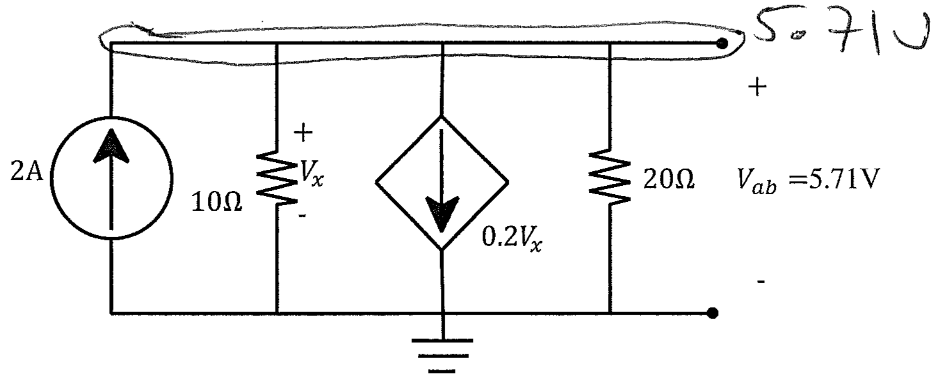
Vraag 1.5 [2]	Question 1.5 [2]
Bereken die aantal nodes en takke in die stroombaan.	Determine the number of nodes and branches in the circuit.



5 nodes ✓

7 branches ✓

Vraag 1.6 [2]	Question 1.6 [2]
Bepaal die drywing ge-assosieer met die afhanklike stroombron.	Find the power associated with the dependent current source.

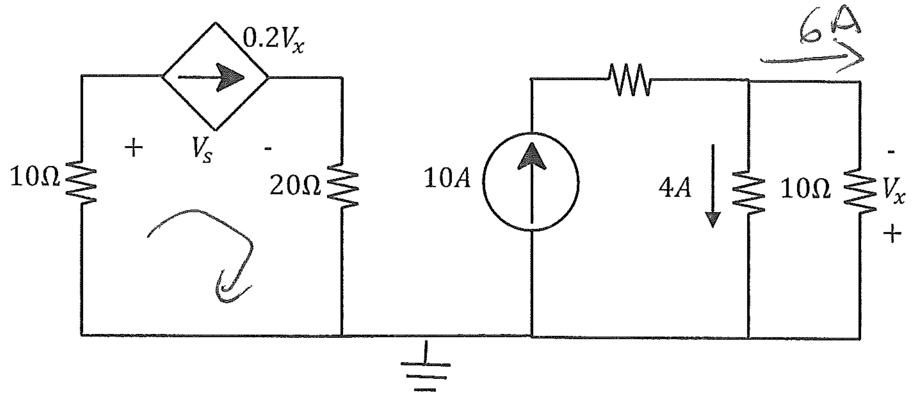


$V_x = 5.71\ V$

$I = 0.2 \times 5.71$
$= 1.142\ A$

$\therefore P = +(V \times I)$
$= 5.71 \times 1.142$
$= \underline{6.52\ W}$ ✓✓

Vraag 1.7 [2]	Question 1.7 [2]
Bepaal die spanning oor die afhanklike stroombron, V_s.	Find the voltage across the dependent current source, V_s.



$V_x = -6 \times 10$
$= -60\,V$

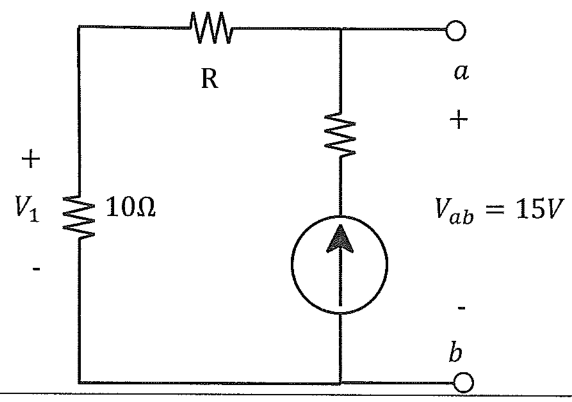
KVL

$(0.2V_x)10 + V_s + (0.2V_x)20 = 0$

$-120 + V_s - 240 = 0$

$\therefore V_s = 360\,V$

Vraag 1.8 [2]	Question 1.8 [2]
Bepaal die waarde van *R*, sodat $V_1 = 5$V.	Find the value of *R*, such that $V_1 = 5$V.



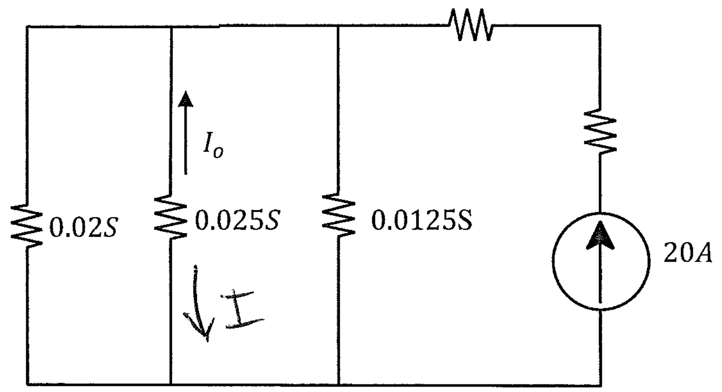
Voltage Division

$V_1 = \left(\frac{10}{10+R}\right) 15 = 5$

$\therefore \left(\frac{10}{10+R}\right) = \frac{5}{15}$

$\therefore R = 20\,\Omega$

Vraag 1.9 [2] Bepaal die stroom I_o.	**Question 1.9 [2]** Find the current I_o.

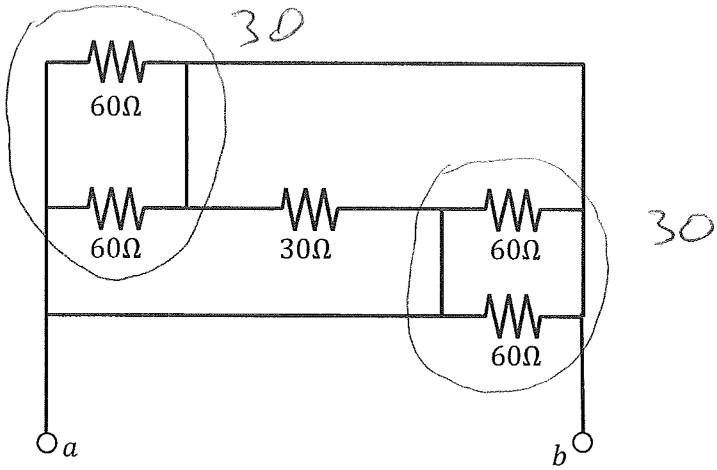


$$I = \left(\frac{0.025}{0.02 + 0.025 + 0.0125}\right) 20$$

$$= 8.7\ A$$

$$\therefore\ I_o = -I = \underline{\underline{-8.7\ A}}$$

Vraag 1.10a [2]	Question 1.10a [2]
Bepaal die ekwivalente weerstand tov terminale a-b.	Find the equivalent resistance wrt terminals a-b.



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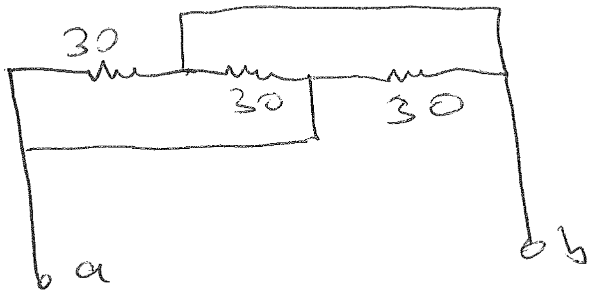

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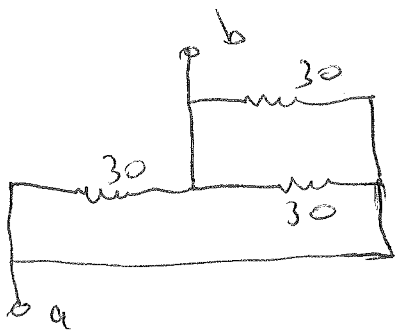

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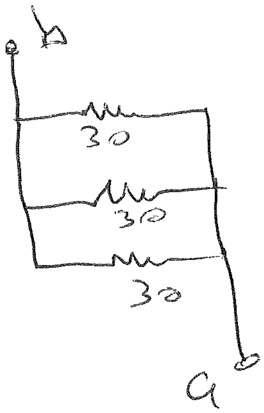
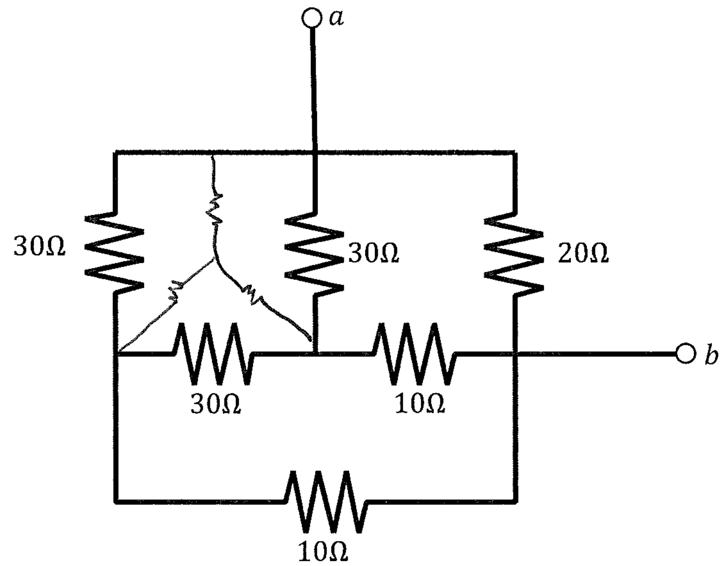
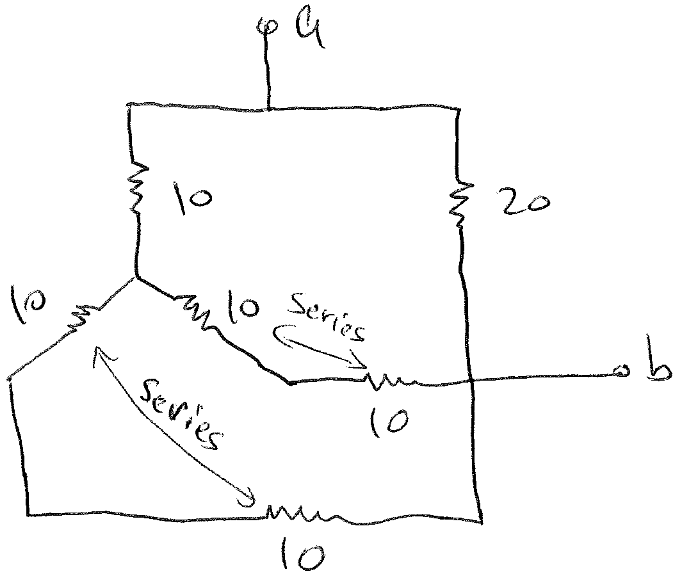


$\therefore R_{eq} = 10\,\Omega$ //

Vraag 1.10b [2] Bepaal die ekwivalente weerstand tov terminale a-b.	**Question 1.10b [2]** Find the equivalent resistance wrt terminals a-b.





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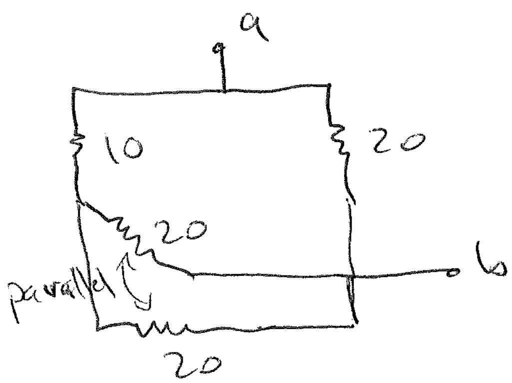
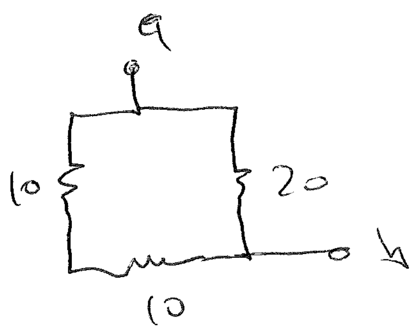


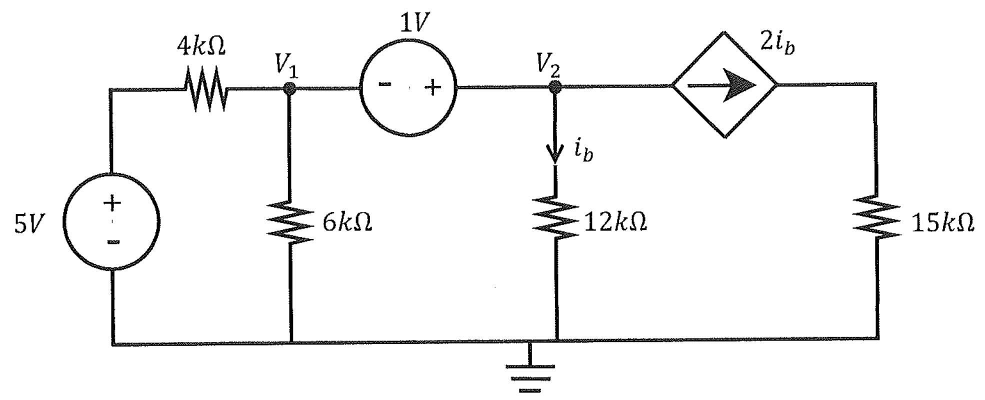
$\simeq$



$\therefore R_{eq} = 10\,\Omega$ //

VRAAG/QUESTION 2 [23]

Vraag 2.1 [4]	**Question 2.1 [4]**
Gebruik node-analise om twee vergelykings van die volgende vorm op te stel: $$aV_1 + bV_2 = -1$$ $$5V_1 + cV_2 = d \quad \text{(KCL)}$$	Use nodal analysis to set up two equations of the following form: $$aV_1 + bV_2 = -1$$ $$5V_1 + cV_2 = d \quad \text{(KCL)}$$



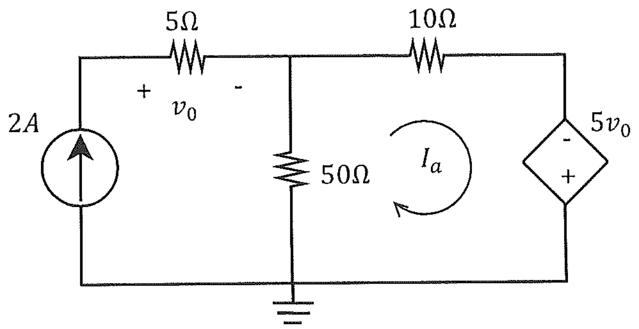
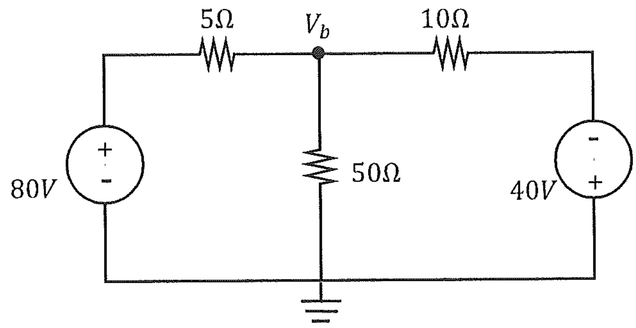
$V_1 + 1 = V_2$

$\Rightarrow \quad V_1 - V_2 = -1$ ✓✓

$$\frac{V_1 - 5}{4000} + \frac{V_1}{6000} + \frac{V_2}{12\,000} + 2 \cdot \frac{V_2}{12\,000} = 0$$

$\times 12\,000$: $\quad 3V_1 - 15 + 2V_1 + V_2 + 2V_2 = 0$

$5V_1 + 3V_2 = 15$ ✓✓

Vraag 2.2 [4] Bereken I_a en V_b.	**Question 2.2 [4]** Determine I_a and V_b.
	

$$10I_a - 5V_0 + 50(I_a - 2) = 0$$

$$V_0 = 2 \times 5 = 10V$$

$$10I_a - 50 + 50I_a - 100 = 0$$

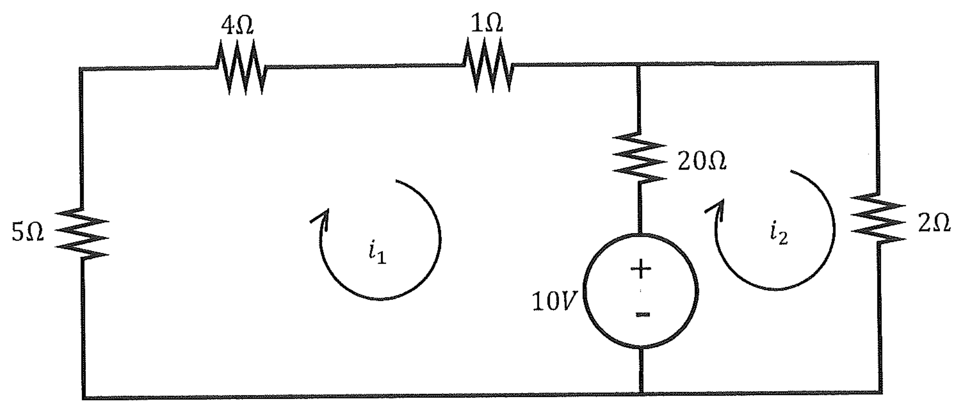
$$60I_a = 150 \Rightarrow I_a = 2.5A \checkmark\checkmark$$

$$\frac{V_b - 80}{5} + \frac{V_b}{50} + \frac{V_b - (-40)}{10} = 0$$

$$\times 50: \quad 10V_b - 800 + V_b + 5V_b + 200 = 0$$

$$16V_b = 600 \Rightarrow V_b = 37.5V \checkmark\checkmark$$

Vraag 2.3 [4]	Question 2.3 [4]
Gebruik lus-analise om twee vergelykings van die volgende vorm op te stel: $30i_1 + ei_2 = f$; $gi_1 + hi_2 = 10$	Use mesh analysis to set up two equations of the following form: $30i_1 + ei_2 = f$; $gi_1 + hi_2 = 10$



$$10i_1 + 20(i_1 - i_2) + 10 = 0$$

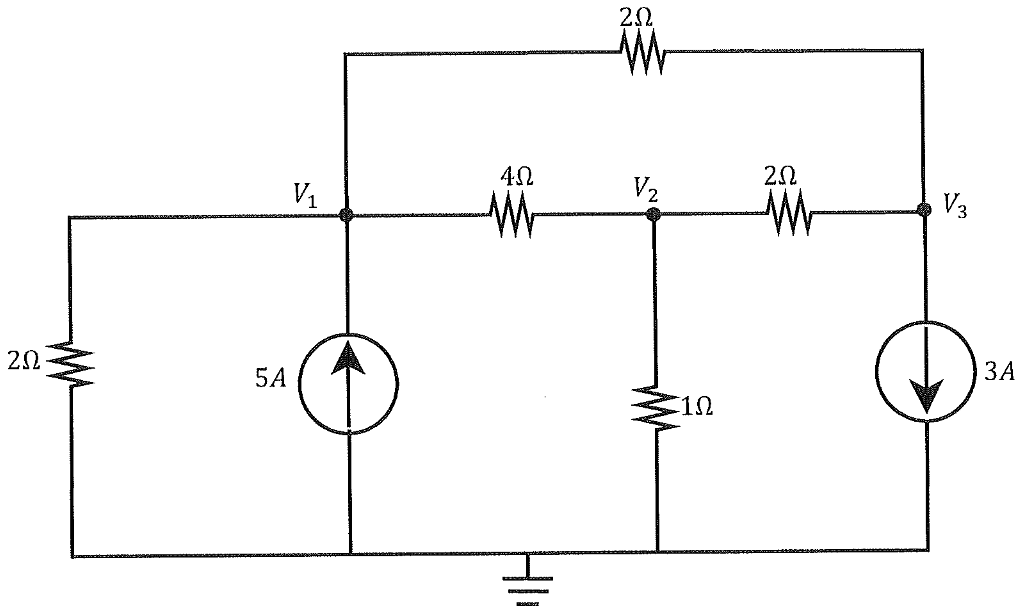
$$30i_1 - 20i_2 = -10$$

$$2i_2 - 10 + 20(i_2 - i_1) = 0$$

$$2i_2 + 20i_2 - 20i_1 = 10$$

$$-20i_1 + 22i_2 = 10$$

Vraag 2.4 [3]	Question 2.4 [3]
Stel `n matriksvergelyking d.m.v. inspeksie op in terme van die nodespannings.	Use inspection to set up a matrix equation in terms of the node voltages.



$$\begin{bmatrix} 1.25 & -0.25 & -0.5 \\ -0.25 & 1.75 & -0.5 \\ -0.5 & -0.5 & 1 \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \\ V_3 \end{bmatrix} = \begin{bmatrix} 5 \\ 0 \\ -3 \end{bmatrix}$$

Vraag 2.5 [4]

a) Bepaal i_2:

$$5i_1 - 10i_2 = 20$$

$$i_1 + i_2 = 2$$

b) Bepaal V_2:

$$\begin{bmatrix} 5 & 0 & -2 \\ 0 & 2 & 0 \\ -2 & 0 & 1 \end{bmatrix}\begin{bmatrix} V_1 \\ V_2 \\ V_3 \end{bmatrix} = \begin{bmatrix} 0 \\ -2 \\ 0 \end{bmatrix}$$

Question 2.5 [4]

a) Determine i_2:

$$5i_1 - 10i_2 = 20$$

$$i_1 + i_2 = 2$$

b) Determine V_2:

$$\begin{bmatrix} 5 & 0 & -2 \\ 0 & 2 & 0 \\ -2 & 0 & 1 \end{bmatrix}\begin{bmatrix} V_1 \\ V_2 \\ V_3 \end{bmatrix} = \begin{bmatrix} 0 \\ -2 \\ 0 \end{bmatrix}$$

a)

$$i_1 = 2 - i_2$$

$$5(2 - i_2) - 10i_2 = 20$$

$$10 - 5i_2 - 10i_2 = 20$$

$$-15i_2 = 10 \Rightarrow i_2 = -0.67 \checkmark\checkmark$$

b)

$$\Delta = \begin{vmatrix} 5 & 0 & -2 \\ 0 & 2 & 0 \\ -2 & 0 & 1 \end{vmatrix} = 10 - 8 = 2$$

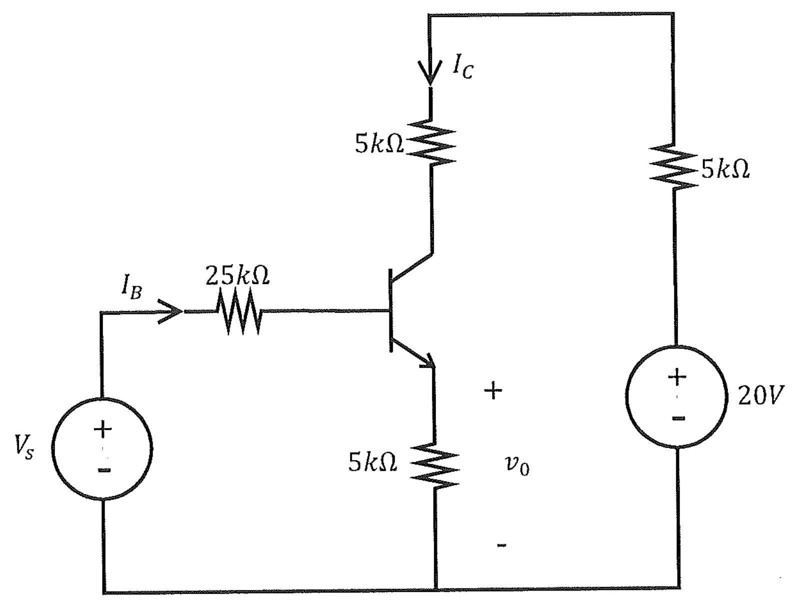
$$\begin{matrix} 5 & 0 & -2 \\ 0 & 2 & 0 \end{matrix}$$

$$\Delta_2 = \begin{vmatrix} 5 & 0 & -2 \\ 0 & -2 & 0 \\ -2 & 0 & 1 \end{vmatrix} = -10 - (-8) = -2$$

$$\begin{matrix} 5 & 0 & -2 \\ 0 & -2 & 0 \end{matrix}$$

$$V_2 = -1 \checkmark\checkmark$$

Vraag 2.6 [4]	Question 2.6 [4]
Gegee: $\beta = 100$; $V_{BE} = 0.7$ V. a) Indien $V_S = 10$ V, bepaal I_B (in μA). b) Indien $I_B = 5$ μA, bepaal v_O.	Given: $\beta = 100$; $V_{BE} = 0.7$ V. a) If $V_S = 10$ V, determine I_B (in μA). b) If $I_B = 5$ μA, determine v_O.



a) $-10 + 25000 I_B + 0.7 + 5000(101 \times I_B) = 0$

$$I_B = \frac{9.3}{25000 + 505000} = 17.55 \mu A$$ ✓✓

b) $V_0 = I_E \times 5000$

$= 101 \times 5 \times 10^{-6} \times 5000$

$= 2.525 V$ ✓✓